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SYSTEMS AND METHODS FOR DATA PRIORITIZATION, DISTRIBUTION, AND MANAGEMENT

Abstract

Disclosed herein are system, method, and computer program product embodiments for intelligent data retrieval, caching, and display. In some embodiments, the system generates a (GUI) in response to a request from a user device. The system may receive the location of the user device and identification of an initiative from the user device. The system may identify a metric correlated with the initiative based on an analysis of previously received initiatives. The system may identify and retrieve a value of the metric from a data source storing a plurality of metrics. The system may store the metric value at a local data store closer in proximity to the user device than the plurality of metrics at the data source. The system may then generate an updated graphical user interface (GUI) with the initiative, metric, and metric value.

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Background/Summary

BACKGROUND

[0001] In enterprise computing environments, data may be distributed across different enterprise applications, data repositories, and physical locations, and may be accessed by devices from different physical locations. However, in instances where a large amount of data is stored, accessing the data may lead to network bottlenecks. For example, if multiple client devices request access to the same data, handling the requests may result in a bottleneck at the data source. The bottleneck may result in lengthy response times or connection timeouts. Furthermore, certain applications (e.g., banking, healthcare) may require high-fidelity and real-time information. In these fields, it is imperative that the results be accessed promptly and reflect the ground truth. Therefore, a method to efficiently retrieve stored data, without causing network bottlenecks, is needed.

[0002] Even if a device is capable of accessing the stored data, it may be difficult to determine how to prioritize what data to retrieve or display. In a big data environment, it is impractical for a human to sift through the data to determine what is important. For example, a database may include millions of data entries. It would be impractical to display every data point within the database. Thus, there is a need to not only efficiently retrieve stored data, but to also identify and display data that is most relevant to a requester. And the need to efficiently identify and display critical information is especially acute when data is accessed on mobile devices, which are typically equipped with small screens that limit the amount of information that can be displayed.

BRIEF SUMMARY

[0003] Disclosed herein are system, apparatus, device, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for intelligent data retrieval, caching, and display. Some embodiments relate to a method comprising: generating a (GUI) in response to a request from a user device, where the request includes a location of the user device; receiving, via the GUI, an interaction comprising an identification of a first metric; identifying a data source comprising the first metric; retrieving a first metric value corresponding to the first metric; storing the first metric value at a first local data store, where the first local data store is at a location closer to the user device location than the data source; and customizing the GUI according to the first metric and the first metric value.

[0004] Some embodiments relate to a system comprising: a memory; and at least one processor coupled to the memory and configured to: generate a (GUI) in response to a request from a user device, where the request includes a location of the user device; receive, via the GUI, an interaction comprising an identification of a first metric; identifying a data source comprising the first metric; retrieve a first metric value corresponding to the first metric; store the first metric value at a first local data store, where the first local data store is at a location closer to the user device location than the data source; and customize the GUI according to the first metric and the first metric value.

[0005] Some embodiments relate to a non-transitory computer-readable device having instructions stored thereon. When the instructions are executed by at least one computing device, the instructions cause the at least one computing device to perform operations comprising: generating a (GUI) in response to a request from a user device, where the request includes a location of the user device; receiving, via the GUI, an interaction comprising an identification of a first metric; identifying a data source comprising the first metric; retrieving a first metric value corresponding to the first metric; storing the first metric value at a first local data store, where the first local data store is at a location closer to the user device location than the data source; and customizing the GUI according to the first metric and the first metric value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The accompanying drawings are incorporated herein and form a part of the specification.

[0007] FIG. **1** depicts an exemplary environment for data prioritization, distribution, and management, according to some embodiments.

[0008] FIG. **2** depicts an exemplary environment including multiple local stores, according to some embodiments.

[0009] FIG. **3** depicts an exemplary interface for displaying initiatives, according to some embodiments.

[0010] FIG. **4** depicts an exemplary interface for creating an initiative, according to some embodiments.

[0011] FIG. **5** depicts an exemplary interface for displaying initiative details, according to some embodiments.

[0012] FIG. **6** depicts another exemplary interface for displaying initiative details, according to some embodiments.

[0013] FIG. **7** depicts an exemplary interface for generating a notification for an initiative, according to some embodiments.

[0014] FIG. **8** depicts an exemplary interface for viewing a report, according to some embodiments.

[0015] FIG. **9** is a flowchart illustrating an example process for data prioritization, distribution, and management, according to some embodiments.

[0016] FIG. **10** is a flowchart illustrating an example process for using a batch process, according to some embodiments.

[0017] FIG. **11** is a flowchart illustrating an example process for utilizing device location, according to some embodiments.

[0018] FIG. **12** is a flowchart illustrating an example process for utilizing a target metric value, according to some embodiments.

[0019] FIG. **13** is a flowchart illustrating an example process for adding a second user to an initiative, according to some embodiments.

[0020] FIG. **14** is a flowchart illustrating an example process for using an initiative to identify additional metrics, according to some embodiments.

[0021] FIG. **15** is a flowchart illustrating an example process for displaying multiple metrics, according to some embodiments.

[0022] FIG. **16** depicts an example computer system useful for implementing various embodiments.

[0023] In the drawings, like reference numbers generally indicate identical or similar elements. Additionally, generally, the left-most digit(s) of a reference number identifies the drawing in which the reference number first appears.

DETAILED DESCRIPTION

[0024] Provided herein are system, apparatus, device, method and/or computer program product embodiments, and/or combinations and sub-combinations thereof, for data prioritization, distribution, and management. The embodiments may be configured to retrieve and cache data at locations close to end-user devices. Data may be cached at an end-user device's request, based on a system prediction, or a combination of both. For example, the system may predict, based on previous user device interactions, data that may be useful to a current user device. In response, the system may preemptively retrieve and locally store the predicted data. The system may further update storage locations based on the location of the end-user device. For example, if the system detects the end-user device location has changed, the system may identify a new storage location closer to the end-user device where data relevant to the user may be stored.

[0025] Embodiments may further support an application operated by end-user devices. The end-user application may be operated by a user device to view, manage, and otherwise interact with the cached data. By intelligently determining data to be cached, and storing the selected data in close proximity to users, embodiments enable high fidelity, real-time data access while reducing concurrent database access and avoiding network bottlenecks. Embodiments further enable users to quickly locate and view critical data within a single graphical user interface.

[0026] FIG. **1** depicts an exemplary analytics environment **100** for data prioritization, distribution, and management, according to some embodiments. Analytics environment **100** includes user computing device **110**, network **120**, analytics engine **130**, local store **134**, data provider systems **140-1**, **140-2**, and **140-3**, and data store **142**.

[0027] User computing device **110** may be any entity attempting to communicate with analytics engine **130**. Although a single user computing device **110** is depicted for purposes of discussion, analytics environment **100** may include multiple user computing devices **110**. User computing device **110** may be a computer system such as computer system **1600** described with reference to FIG. **16**. User computing device **110** may be a client system such as a desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, and/or other computing device capable of communicating over network **120**.

[0028] User computing device **110** includes communication interface **112-1**, display engine **114**, and user application **116**. Communication interface **112-1** may be configured to communicate with analytics engine **130** via network **120**. Communication interface **112-1** may comprise any suitable network interface capable of transmitting and receiving data, such as, for example a modem, an Ethernet card, a communications port, or the like. Communication interface **112-1** may be able to transmit data using any wireless transmission standard such as, for example, Wi-Fi, Bluetooth, cellular, or any other suitable wireless transmission. Display engine **114** may be configured to display information at user computing device **110**. Display engine **114** may be configured to receive interactions from a user. An interaction may refer to various forms of user input, for example, a click, a button press, a swipe, etc. User application **116** may interact with analytics engine **130** to retrieve and display data stored in local store **134**. In some embodiments, user application **116** may be a native application installed on user computing device **110**. In some embodiments, user application **116** may be a web application accessed by user computing device **110**. In such a case, the web application may be hosted by analytics engine **130** or on a separate server connected to network **120**. A user may access, view, and interact with user application **116** via display engine **114**.

[0029] In some embodiments, user computing device **110** may use user application **116** to define initiatives, which may be transmitted to analytics engine **130**. In a big data environment where large quantities of data are distributed across multiple data sources, such as many enterprise environments, it is impractical to manually sift through the data to identify useful information. To aid the identification of information useful to particular users, embodiments enable initiatives to be defined. The initiative may be used to describe a problem or goal of the user or an organization with which the user is associated. For example, an initiative may be defined to increase network throughput or decrease network latency. As another example, an initiative may be defined to increase the efficiency of an organizational process or workflow. As yet another example, an initiative may be defined to increase revenue or profitability. In some embodiments, initiatives may be automatically defined or suggested by analytics engine **130** based on various factors, such as user roles, past user activity, or other initiatives defined by the user or organization. Analytics engine **130** may suggest additional initiatives based on attributes associated with existing initiatives. For example, user computing device **110** may define an initiative to reduce costs associated with a hospital's psychiatric department. Based on the initiative, analytics engine **130** may be configured to recommend an additional initiative to reduce costs associated with the hospital's surgery department.

[0030] User computing device **110** may further define data or metrics associated with an initiative. As will be discussed further below, analytics engine **130** may receive the initiative, determine relevant data metrics, and store the data associated with each metric in close proximity to user computing device **110**. This configuration: (1) decreases network bottlenecks by distributing data to local data stores, reducing the number of user computing devices **110** accessing the same data store; (2) increases access speed by staging data near user computing device **110**; and (3) analytics engine **130** may use previous interactions from other user computing devices **110** to suggest and preemptively retrieve additional metrics. User computing device **110** may communicate with analytics engine **130** via network **120**.

[0031] Network **120** may be any type of computer or telecommunications network capable of communicating data, for example, a local area network, a wide-area network (e.g., the Internet), or any combination thereof. The network may include wired and/or wireless segments. Network **120** may also represent or include segments corresponding to a secure enterprise network.

[0032] Analytics environment **100** may also include one or more data provider systems **140**. For example, analytics environment **100** includes data provider systems **140-1**, **140-2**, and **140-3**. Each data provider system **140** may have and manage data within its own data store **142**.

[0033] Each data store **142** may be implemented using a memory storage device. Each data store **142** may be configured to store various data associated with the corresponding data provider system **140**, which may be accessed by analytics engine **130**. Data store **142** may be organized according to various schemes and may employ one or more databases for storing and organizing data. Data provider system **140** may use communications interface **112** to access network **120**.

[0034] As depicted, data within analytics environment **100** may be disparate and fragmented across multiple data provider systems **140**. For example, data provider systems **140** may represent different systems within an enterprise organization (e.g., billing system, customer relationship management system, network monitoring system, etc.), or systems associated with different organizations or departments within an organization. By utilizing analytics engine **130**, selective data from each data provider system **140** may be retrieved and stored in close proximity to user computing device **110**. This may reduce bottlenecks at network **120** while providing user computing device **110** real-time data access. Additionally, analytics engine **130** may provide user computing device **110** data it may not have previously attempted to access. For example, user computing device **110** may define an initiative and desire data metrics maintained by data provider system **140-1**. Analytics engine **130** may analyze the initiative and determine additional data relevant to the initiative that is maintained by data provider system **140-2**. This determination may be based on, for example, the result of a machine learning model prediction and/or analysis of initiatives from other user computing devices **110**.

[0035] Analytics engine **130** may be implemented using one or more servers and/or databases. In some embodiments, analytics engine **130** may be implemented using a computing device such as a desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, and/or other computing device. In some embodiments, analytics engine **130** may be implemented as an application in an enterprise computing system and/or a cloud-computing system. In some embodiments, analytics engine **130** may be implemented on a computer system such as computer system **1600** described with reference to FIG. **16**. Although a single analytics engine **130** is depicted, analytics environment **100** may include multiple analytics engines **130**.

[0036] Analytics engine **130** includes query engine **132** and communication interface **112-2**. Analytics engine **130** may be in communication with local store **134**. Local store **134** may be a storage device configured to store data from data stores **142-1**, **142-2**, and **142-3**. Local store **134** may employ one or more databases for storing and organizing data.

[0037] Query engine **132** may be configured to retrieve data from each of data provider systems **140-1**, **140-2**, and **140-3**, and save it to local store **134**. Query engine **132** may communicate, through communication interface **112-2**, with each of data provider systems **140-1**, **140-2**, and **140-3** via network **120**. Query engine **134** may be configured to retrieve certain metrics from a data provider system **140**. As discussed above, user computing device **110** may define and send an initiative to analytics engine **130**. User computing device **110** may also send metrics associated with the initiative. In some embodiments, analytics engine **130** may determine, based on the initiative, additional metrics that query engine **132** should retrieve. The metric may be a type of data. For example, for a healthcare organization, a metric may be average surgery cost per month. A metric may have a corresponding metric value, for example $1,000,000.

[0038] In some embodiments, each time query engine **132** obtains a new metric value, it may overwrite the current value at local store **134**. Such a process is beneficial to store the most up-to-date information as well as reduce the amount of storage local store **134** requires. In some embodiments, query engine **132** may be configured to maintain a history of metric values at local store **134**. For example, a metric value may be updated once per day, and local store **134** may be configured to keep the last thirty metric values. In this embodiment, once thirty metric values are stored at local store **134**, query engine **132** may overwrite the oldest metric value. Storing a limited history of metric values allows local store **134** to remain small while also providing user computing device **110** the benefit of historical data.

[0039] Query engine **134** may be further configured to retrieve metrics and metric values that analytics engine **130** determines are linked to the initiative received from user computing device **110**. In some embodiments, the determination may be made via a historical analysis of previous initiatives. For example, an initiative may involve reducing heart surgery costs. Analytics engine **130** may determine that surgery costs vary based on region. In this example, query engine **132** may be configured to retrieve metric values for average surgery costs in the region local to user computing device **110**.

[0040] In some embodiments, analytics engine **130** may also use artificial intelligence to determine which metrics to retrieve based on the initiative. In some embodiments, analytics engine **130** may use a trained machine learning model to identify which metrics to retrieve. For example, analytics engine **130** may include a list of previously received initiatives, metrics and target metrics (i.e., goals) for each initiative, and the time to reach each target metric. Analytics engine **130** may use this information to determine which metrics may be most effective for each initiative. For instance, analytics engine **130** may have received the same initiative (e.g., reduce orthopedic surgery waste) from two different user computing devices **110**. Each user computing device **110** may have indicated different metrics to track associated with the initiative. For example, the first user computing device **110** may have tracked time spent per surgery, while the second user computing device **110** may have tracked the value of material lost per surgery. Here, the second user computing device **110** may have reached their target metric faster than the first user computing device **110**. Analytics engine **130** may use this information to recommend a metric tracking the value of material lost per surgery upon receiving a new initiative associated with reducing surgical costs.

[0041] Analytics engine **130** may use data associated with existing or past initiatives to train a machine learning model capable of linking metrics and initiatives. Use of such machine-learning techniques may be beneficial to accurately identify data metrics useful for tracking and evaluating the progress of certain initiatives, which often may not be known or obvious to users defining initiatives. Analytics engine **130** may additionally use reinforcement learning to update a machine learning model. Reinforcement learning may involve receiving scores or feedback from user computing device **110** in response to analytics engine's **130** actions. For example, analytics engine **130** may recommend additional metrics to user computing device **110** based on the output of a machine learning model. User computing device **110** may send a score for each recommended metric to analytics engine **130**. Analytics engine **130** may re-train the machine learning model while taking the received scores into account. This feedback may be beneficial to improve analytics engine's **130** ability to recommend metrics based on initiatives.

[0042] In some embodiments, query engine **132** may retrieve data according to an update frequency or schedule. For example, query engine **132** may be configured to retrieve updated metric values from data stores **142** once a week, once a day, once an hour, etc. User computing device **110** may set the update frequency. Analytics engine **130** may be configured with a default update frequency (e.g., once per day). Query engine **132** may have an update frequency for each metric that it is configured to retrieve from data stores **142**. Certain metrics may have higher priority than others and may be updated at higher frequencies. Analytics engine **130** may configure query engine **132** to execute a batch process. The batch process may include one or more metrics and an update frequency. Query engine **132** may execute the batch process, according to the update frequency, to retrieve metric values corresponding to the metrics. Metric values may be stored at local store **134**.

[0043] Query engine **132** may retrieve data from multiple data provider systems **140**. For example, query engine **132** may retrieve data from data provider system **140-1**, **140-2**, and **140-3**. In some embodiments, data may be formatted differently across multiple data provider systems **140**. Analytics engine **130** may be configured to reformat the received data so that each metric has the same format. For example, data provider system **140-1** may store average surgery cost per year and data provider system **140-2** may store average surgery cost per month. Here, analytics engine **130** may retrieve data from both data provider systems **140** but reformat data provider system's **140-2** data. For example, analytics engine **130** may multiply data provider system's **140-2** value by twelve so that both may be stored as cost per year. Such reformatting may be beneficial so that the received metrics and metric values may be consistent, compared, or aggregated. In some embodiments, analytics engine **130** may reformat the data prior to storing it at local store **134**. In some embodiments, analytics engine **130** may reformat the data after it has been stored at local store **134**.

[0044] In some embodiments, local store **134** may be in closer proximity to user computing device **110** than data provider systems **140**. Such an architecture may have multiple benefits. First, because local store **134** is closer in proximity to user computing device **110** than data provider systems **140**, the amount of time for user computing device **110** to access data is decreased. Local store **134** may also be configured to serve only users within a certain geographic proximity, limiting network traffic and reducing bottlenecks both at local store **134** and on network **120**. Additionally, local store **134** may include only a subset of maintained by data provider systems **140**, leading to faster data retrieval times. Furthermore, user computing device **110** may be a device with a small screen such as a cell phone. Therefore, it may be difficult for user computing device **110** to effectively visualize and interact with a large amount of data. By storing a subset of data from data provider systems **140** at local store **134**, user computing device **110** may be able to effectively view and interact with the subset of data, even on a device such as a cell phone.

[0045] Analytics engine **130** may further be configured to communicate with user computing device **110**. For example, analytics engine **130** may be further configured to send notifications or alerts. Notifications may be sent as SMS messages, emails, and/or phone calls. Notifications may also be sent through an application (e.g., user application **116**) installed on or accessible from user computing device **110**. The notification may include a link that, when interacted with, launches a GUI providing an interface to analytics engine **130**. In some embodiments, analytics engine **130** may send a notification to user computing device **110** when a metric value reaches a target metric value. In some embodiments, analytics engine **130** may send a notification to user computing device **110** when a metric value differs from a target metric value by a predefined threshold. For example, a metric may be monthly revenue and a target metric value may be set to $1,000,000. In this example, lower and upper threshold values may be set at $800,000 and $1,200,000, respectively. If analytics engine **130** detects that a metric value for monthly revenue falls outside of these lower and upper thresholds, analytics engine **130** may send a notification to user computing device **110**. In some embodiments, an alert may launch user application **116** (if not already executing) and cause user application **116** to show the metric (e.g., monthly revenue), metric value (e.g., $700k), and target metric value (e.g., $1 million).

[0046] Analytics engine **130** may be further configured to automatically send updated metric values to user computing device **110**. In some embodiments, user computing device **110** may require rapid access to its initiatives along with the initiative's metrics and metric values. In order to provide this on-demand access, analytics engine **130** may send updated metrics values to user application **116** at user computing device **110**. As a result, a user interacting with user application **116** may be able to view the most up to date information, without having to wait for new metric values to be retrieved. The frequency that analytics engine **130** sends updates to user computing device **110** may vary. In some embodiments, analytics engine **130** may update user computing device **110** each time query engine **132** obtains new metric values. In some embodiments, user computing device **110** sets a frequency at which analytics engine **130** sends updated metric values.

[0047] Analytics engine **130** may add or associate additional users with current initiatives managed by analytics engine **130**. Analytics engine **130** may automatically add users or add them based on a request. For example, a first user may send a request to analytics engine **130** to add a second user to an initiative. The message may be sent via user computing device **110**. In response, analytics engine **130** may send a notification to user computing devices **110** associated with each user. The notification may state that the second user has been added to the initiative. Analytics engine **130** may automatically add or associate a user with an initiative based on an analysis of the initiative. For example, a user may define an initiative to reduce orthopedic surgery costs. In response, analytics engine **130** may be configured to add each orthopedic surgeon to the initiative without a request from the user.

[0048] Analytics engine **130** may further allow user computing devices **110** to send communications to other user computing devices **110**. For example, a first user computing device **110** may access analytics engine **130** via user application **116** and send a message to a second user computing device **110**. Messaging via analytics engine **130** may reduce network latency since metrics of interest may already be stored near analytics engine **130** at local store **134**. Therefore, there may be little to no delay associated with retrieving metrics to send. In some embodiments, user application **116** may be configured to allow a user computing device **110** to send a link to a specific part of user application **116**. For example, a metric may have fallen below a certain threshold and a first user computing device **110** may wish to communicate this to a second user computing device **110**. In such a case, the first user computing device **110** may send a link, via user application **116**, to a second user computing device **110**. Once accessed, the link may cause second user computing device's **110** user application **116** to display the initiative, metric, and metric value.

[0049] Analytics engine **130** may associate a user operating computing device **110** with a permission scope or level. The permission scope may be used by analytics engine **130** to determine what actions user computing device **110** may take. Permissions may be defined in various manners. In some embodiments, permissions may be automatically defined according to users' roles within an organization or based on other known characteristics of users. In some embodiments, permissions may be manually defined by an administrator. For example, a team leader within an organization may be able to create new initiatives, edit existing initiatives, add metrics to existing initiatives, delete metrics from existing initiatives, view metric values, add and remove team members from initiatives, and set permissions for team members. In some embodiments, permissions may be assigned for individual initiatives. For example, a team leader may allow a first team member to add metrics to an initiative, but prevent a second team member from adding metrics to that same initiative.

[0050] FIG. **2** depicts an exemplary environment **200** with multiple local stores **134**, according to some embodiments. Analytics environment **200** includes analytics engine **130**, user computing device **110**, network **120**, local stores **134-1** and **134-2**, and data provider system **140**. Although a single data provider system **140** is depicted, environment **200** may include multiple data provider systems **140**. Analytics engine **130** may store data at local store **134-1** or **134-2** based on the distance or latency time between each local store **134** and user computing device **110**. For example, distance **210-1** between user computing device **110** and local store **134-1** engine **130-1** may be less than distance **210-2** between user computing device **110** and local store **134-2**. Here, metrics for user computing device **110** (e.g., metrics linked to initiatives associated with the user operating user computing device **110**) may be stored at local store **134-1** instead of local store **134-2**. This architecture may reduce network bottlenecks with multiple user devices **110** accessing a single local store **134**. Additionally, staging metrics closer to the location of user device **110** reduces latency in providing critical metrics to user device **110**.

[0051] Analytics engine **130** may update local store **134** based on user computing device **110** moving to an updated location. For example, user computing device **110** may move closer to local store **134-2**. In response, analytics engine **130** may save data to local store **134-2** instead of local store **134-1**. This may be beneficial to ensure that user computing device **110** has rapid access to data at local store **134**. Analytics engine **130** may determine the location of user computing device **110** in various ways. In some embodiments, analytics engine **130** may request the location of user computing device **110**. In some embodiments, analytics engine **130** may command each local store **134** to send a ping and measure the response time from user computing device **110**. Analytics engine **130** may use the response times to determine which local store **134** is closest to user computing device **110**.

[0052] FIG. **3** depicts an exemplary interface **300** for displaying initiatives, according to some embodiments. User computing device **110** may access interface **300** when it connects to analytics engine **130** via user application **116**. Interface **300** includes initiative **310**, metric **320**, metric value **325**, target **330**, target date **340**, view details **350**, notify team **360**, refresh **365**, create alert **370**, edit metrics **380**, and new initiative **390**. In some embodiments, user application **116** may be configured to automatically load interface **300** upon initiation of user application **116**. Such operation enables the user operating user application **116** quick access to critical information relevant to the user, as discussed further below.

[0053] As depicted, interface **300** may display multiple initiatives, **310-1** and **310-2**. Metric **320** may be a metric associated with initiative **310**. Metric **320** may be a metric that analytics engine **130** may retrieve from data provider system **140** and store at local store **134**. For example, metric **320** may be accounts receivable. Metric value **325** may be the value corresponding to metric **320**. For example, metric **320** may be accounts receivable and metric value **325** may be $1 million. Target **330** may be a goal corresponding to metric **320**. For example, target **330**, corresponding to metric **320**, may be $1 million. Target date **340** may be a date that user computing device **110** wishes metric **320** to reach target **330** by. Target date **340** may include a date and/or time. Initiative **310-1** and **310-2** may include multiple metrics **320**, metric values **325**, targets **330**, and target dates **340**.

[0054] View details **350** may enable user computing device **110** to view additional information regarding initiative **310**. For example, initiative **310** may have multiple metrics **320** associated with it and interface **300** may display a limited set. User computing device **110** may identify which metrics **320** to display on interface **300**. In some embodiments, analytics engine **130** may identify which metrics **320** to display. In some embodiments, analytics engine **130** may configure interface **300** to display a metric **320** with a metric value **325** that changed the most over a certain time period. In some embodiments, analytics engine **130** may configure interface **300** to display a metric **320** that is determined to have the most impact on initiative **310**. In some embodiments, multiple metrics may be displayed for each initiative. In some embodiments, the number of metrics displayed may depend on available display real estate. In some embodiments, view details **350** may enable user computing device **110** to view each metric (or a larger subset of metrics) associated with the initiative stored at local store **134**.

[0055] Displaying a limited set of metrics **320** may be advantageous. As stated above, in a big data environment, analytics engine **130** may have access to thousands or even millions of metrics. A large number of metrics may be tied to a single initiative. Even though these metrics may be stored at local store **134** for rapid access, it is impractical to display all of these metrics in interface **300**. Additionally, user computing device **110** may be a device with a small screen (i.e., a cell phone) and therefore may not be able to display, in a useful manner, every metric. Therefore, analytics engine **130** may configure interface **300** to display the most important metrics **320**, from local store **134**, tied to each initiative **310**. However, view details **350** may enable user computing device **110** to access additional other metrics **320** associated with initiative **310**. For example, view details **350** may display an interface to view all (or a greater subset of) metrics stored at local store **134** associated with initiative **310**. Interacting with view details **350** may launch a new GUI.

[0056] Notify team **360** may be used to send a notification or alert to users associated with initiative **310**. Create alert **370** may launch a new GUI screen to create an alert that will generate and send an alert or notification to user computing devices **110** associated with initiative **310**. The alert may also be automatically generated based on an event occurring (e.g., a metric passing a threshold). Edit metrics **380** may be used to edit metrics currently linked to initiative **310**. For example, user computing device **110** may use edit metrics **380** to launch a new GUI screen to remove current metrics **320** and/or add new metrics **320** to initiative **310**. For example, initiative **310** may be associated with reducing surgical costs, and have metric **320-1** associated with average surgery time. User computing device may click edit metrics **380**, generating a new GUI, to add metric **320-2** associated with material lost per surgery. New initiative **390** may generate a new screen allowing user computing device **110** to create a new initiative **310**.

[0057] FIG. **4** depicts an exemplary interface **400** for creating an initiative, according to some embodiments. Interface **400** may be displayed as a result of user computing device **110** selecting new initiative **390**. Interface **400** may be displayed on a computing device, such as user computing device **110**, when accessing analytics engine **130**. Interface **400** may include various fields, such as name **410**, metric **420**, target **430**, target date **440**, users **450**, and notifications **460**. As stated above, user computing device **110** may define an initiative for analytics engine **130** to retrieve metrics for. Name **410** may be a name to identify the initiative. Metric **420** may be a metric that analytics engine **130** may retrieve from data provider system **140**. For example, metric **420** may be total claimants. Target **430** may be a goal corresponding to metric **420**. For example, metric **420** may value of material wasted during surgical procedures and target **430** may be $0. Target date **440** may be a date by which user computing device **110** wishes metric **420** to reach target **430**. Target date **440** may include a date and/or time.

[0058] Users **450** may correspond to other user computing devices **110** that may access the initiative created via interface **400**. User computing device **110** may identify other user computing devices **110** by listing identifiers (e.g., email address, phone number, employee ID). User computing device **110** may set permissions associated with each identifier. As discussed above, permissions may be used by analytics engine **130** to determine what actions a user computing device **110** may perform on analytics engine **130**. Once added within users **450**, each user computing device **110** may receive an alert or notification with a link to access interface **300**. Notifications **460** may be a checkbox indicating whether the user computing devices **110** should receive auto-notifications associated with the defined initiative. In some embodiments, notifications may be generated when metric **420** reaches a threshold value, such as target **430**.

[0059] FIG. **5** depicts an exemplary interface **500** for displaying initiative details, according to some embodiments. Interface **500** may be launched when user computing device **110** interacts with view details **350**, as depicted in FIG. **3**. Interface **500** may be used to show additional metrics **320** and metric values **325** associated with an initiative **310**. Interface **500** may display metrics **320** in various formats. For example, certain metrics **320-1** and metrics values **325-1** may be depicted as a graphic. In some embodiments, metrics **320-2** and metrics values **325-2** may be depicted as text.

User computing device **110** may interact with and rearrange metrics **320** and metrics values **325** as depicted in interface **500**.

[0060] As discussed above, metric **320** may include metric values **325** from multiple data provider systems **140**. For example, initiative **310** may relate to increasing hospital profits. Analytics engine **130** may receive profit data from three hospitals (i.e., data provider system **140-1**, **140-2**, and **140-3**). Analytics engine **130** may standardize the profit data from each data provider system **140** so that a single metric value may be displayed at interface **500**. For example, each metric value **325** may be formatted to show profit per month. In some embodiments, interface **500** may be configured to provide user computing device **110** access to each metric value **325** from each data provider system **140**. This may be useful to view disparities for a single metric **320** across multiple data provider systems **140**.

[0061] Interface **500** may include notify team **510**, refresh **520**, and view initiatives **530**. Notify team **510** may generate an alert to each user computing device **110** linked to initiative **310**. The alert may include the metrics **320** and metric values **325** at interface **500**. Refresh **520** may retrieve updated metric values **325** from analytics engine **130**. View initiatives **530** may update the GUI to show interface **300**.

[0062] FIG. **6** depicts another exemplary interface **500-2** for displaying initiative details, according to some embodiments. Interface **500-2** may be displayed when analytics engine **130** is accessed via user computing device **110** with a smaller screen (e.g., a cell phone). As a result, analytics engine **130** may prioritize what data to display. For example, if user computing device **110** has access to two initiatives **310-1** and **310-2**, and initiative **310-1** has a metric **320-1** below a predefined threshold, analytics engine **130** may display initiative **310-1**, metric **320-1**, and metric value **325-1**. In some embodiments, user computing device **110** may select an initiative **310**, metric **320**, and metric value **325** to always display. In some embodiments, analytics engine **130** may be configured to display initiative **310**, metric **320**, and metric value **325** that requires attention. For example, metric **325** may have passed a threshold value, causing an alert to be sent to user computing device **110**. If user computing device **110** interacts with the link in the alert to access analytics engine **130**, the link may cause interface **500-2** to be displayed. Interface **500-2** therefore enables a user to quickly access critical information without the need to sift through large amounts of data stored in data provider systems **140**.

[0063] FIG. **7** depicts an exemplary interface **700** for generating a notification for an initiative, according to some embodiments. Interface **700** may be generated when user computing device **110** accesses notify team **510**, as depicted in FIG. **5**. Interface **700** includes message **710** and recipients **720**. User computing device **110** may enter message **710** that will be received by recipients **720**. Message **710** may include any input that user computing device **110** may transmit (e.g., text, audio, photograph, video). Within recipients **720**, user computing device **110** may list identifiers associated with other user computing devices **110** designated to receive message **710**. An identifier may be a name, phone number, employee ID, and/or email address. Analytics engine **130** may send the notification to each recipient listed via network **120**.

[0064] FIG. **8** depicts an exemplary interface **800** for viewing a report, according to some embodiments. Interface **800** may display report **810**. Report **810** may be depicted as a table including each initiative. Each initiative may include metrics associated with the initiative, current metric values, target metric values, and trend lines. Report **810** may be accessed by user computing device **110** connecting to analytics engine **130**. In some embodiments, analytics engine **130** may be configured to generate and send report **810** to user computing device **110**. Analytics engine **130** may generate and send report **810** to user computing device **110** at a predefined interval. In some embodiments, user computing device **110** may request that analytics engine **130** generate and send report **810**. Refresh **820** may be a button configured to update report **810**. For example, when a user selects refresh **820** user computing device **110**, analytics engine **130** may retrieve updated metric values from data store **142**, store them at local store **134**, and update report **810** depicted in interface **800**.

[0065] FIG. **9** is a flowchart illustrating an example process **900** for data prioritization, distribution, and management, according to some embodiments. Process **900** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **9** may be performed by analytics engine **130** and/or user computing device **110**. In some embodiments, one or more of the steps shown in FIG. **9** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **9**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **9**. The steps shown in FIG. **9** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **9**.

[0066] At **910**, analytics engine **130** generates a GUI in response to a request from a user computing device **110**, where the request includes a location of user computer device **110**. For example, user computing device **110** may connect to analytics engine **130**. Analytics engine **130** may generate and cause to be displayed a GUI at user computing device **110**. User computing device **110** may send its location to analytics engine **130**. In some embodiments, analytics engine **130** may request user computing device's **110** location.

[0067] At **920**, an interaction, via the GUI, is received comprising an identification of a first metric. The first metric may be a data point that user computing device **110** desires to track. As non-limiting examples, the first metric may be revenue per month or surgical waste measured over a period of time.

[0068] At **930**, analytics engine **130** identifies a data source comprising the first metric. For example, analytics engine **130** may determine that a data provider system **140** includes the first metric received from user computing device **110**. In some embodiments, analytics engine **130** may query the data provider system **140** to determine it includes the first metric. In some embodiments, analytics engine **130** may include a list including metrics and their respective data provider systems **140** on network **120**.

[0069] At **940**, analytics engine **130** retrieves a first metric value corresponding to the first metric. The first metric value may be a value corresponding to the first metric. For example, if the first metric is revenue per month, the first metric value may be $1,000,000. Analytics engine **130** may use query engine **132** to retrieve the first metric value. Analytics engine **130** may store the first metric value at local store **134**.

[0070] At **950**, analytics engine **130** may store the first metric value at a first local data store, wherein the first local data store is at a location closer to the user device location than the data source. The first local data store may be local store **134**. Analytics engine **130** may identify the first local data store by determining the closest local store **134** to user computing device **110**.

[0071] At **960**, analytics engine **130** may customize the GUI according to the first metric and the first metric value. For example, the GUI on user computing device **110** may be customized to display the first metric and the first metric value retrieved by analytics engine **130**. Analytics engine **130** may configure the GUI in various formats. For example, if analytics engine **130** is configured to store a single value for the first metric, it may display the first metric and value. In some embodiments, analytics engine **130** may be configured to store historical values for the first metric. In some embodiments, analytics engine **130** may cause the GUI to display a graph or plot of the first metric values.

[0072] FIG. **10** is a flowchart illustrating an example process **1000** for using a batch process, according to some embodiments. Process **1000** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **10** may be performed by analytics engine **130** and/or user computing device **110**. In some embodiments, one or more of the steps shown in FIG. **10** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **10**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **10**. The steps shown in FIG. **10** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **10**.

[0073] At **1010**, analytics engine **130** configures a batch process to retrieve a first metric value. The batch process may include a first metric and an update frequency. The batch process may be used to automatically update data at local store **134**. Analytics engine **130** may determine the first metric and update frequency. In some embodiments, user computing device **110** may set the first metric and update frequency.

[0074] At **1020**, analytics engine executes the batch process according to the update frequency. In some embodiments, query engine **132** may retrieve the first metric value from data provider system **140**. At **1030**, analytics engine updates the first metric value at the first local data store. The first local data store may be local store **134**. Utilizing a batch process may be beneficial to ensure that local store **134** has the latest data according to the update frequency.

[0075] FIG. **11** is a flowchart illustrating an example process **1100** for utilizing device location, according to some embodiments. Process **1100** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **11** may be performed by analytics engine **130** and/or user computing device **110**. In some embodiments, one or more of the steps shown in FIG. **11** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **11**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **11**. The steps shown in FIG. **11** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **11**.

[0076] At **1110**, analytics engine **130** detects the user computing device **110** has moved from a location to an updated location. Analytics device **130** may request user computing device's **110** location. User computing device **110** may be configured to automatically send its location to analytics engine **130** when it changes.

[0077] At **1120**, analytics engine **130** may identify a second local data store that is at a location closer to the updated user device location than the first local data store. The second local data store may be identified by measuring the distance or response time between user computing device **110** and each local data store **134**.

[0078] At **1130**, analytics engine **130** stores the first metric value at the second local data store. Using the second local data store that is closer to user computing device **110** is beneficial to reduce latency in providing data to user computing device **110**. Additionally, such an architecture helps to distribute network **120**'s load across multiple entities.

[0079] FIG. **12** is a flowchart illustrating an example process **1200** for utilizing a target metric value, according to some embodiments. Process **1200** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **12** may be performed by analytics engine **130**. In some embodiments, one or more of the steps shown in FIG. **12** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **12**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **12**. The steps shown in FIG. **12** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **12**.

[0080] At **1210**, analytics engine **130** receives a target metric value corresponding to the first metric. The target metric value may be set by user computing device **110**. User computing device **110** may enter the target metric value via the GUI at user application **116**. The target metric value may be used to determine initiative progress. For example, an initiative may be designed to decrease surgical waste and a target metric value may be set at $0 (i.e., no surgical waste).

[0081] At **1220**, analytics engine **130** may calculate a difference between the target metric value and the first metric value. For example, if the first metric is surgical waste, the calculation may be the difference between current surgical waste and no surgical waste. At **1230**, analytics engine **130** generates a report including the first metric, the first metric value, the target metric value, and the difference between the target metric value and the first metric value. In some embodiments, the report may include a graph depicting historical first metric values and the target metric value, allowing users to easily view initiative progress.

[0082] At **1240**, analytics engine **130** sends the report to user computing device **110**. Once received the report may be displayed within the GUI on the user device. Analytics engine **130** may send the report via network **120**. In some embodiments, an alert may be transmitted with the report, which may cause display engine **114** to display user application **116** without any user interaction. This may be useful in a scenario where the report requires immediate attention. In some embodiments, the report may generate a notification (e.g., sound or vibration) on user computing device **110**. In this case, a user may select the notification to display the report on user computing device **110**.

[0083] FIG. **13** is a flowchart illustrating an example process **1300** for adding a second user to an initiative, according to some embodiments. Process **1300** may be used to add users to initiatives and define the level of access or permission they possess with respect to the initiative. For example, additional users may be able to view but not edit metrics within the initiative. Process **1300** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **13** may be performed by analytics engine **130**. In some embodiments, one or more of the steps shown in FIG. **13** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **13**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **13** The steps shown in FIG. **13** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **13**.

[0084] At **1310**, analytics engine **130** receives an identifier associated with a second user computing device **110-2**. The second user computing device **110-2** may be associated with a second user. Analytics engine **130** may receive the identifier from user computing device **110-1**. The identifier may be any value to identify the second user device **110-2**, such as a phone number, email address, or employer ID.

[0085] At **1320**, analytics engine **130** assigns a permission scope to the second user device **110-2**, wherein the permission scope is configured to block an interaction from second user device **110-2** with the GUI. Analytics engine **130** may receive the permission scope from user computing device **110-1** when it identifies second user device **110-2**. The permission scope may be used to determine what interactions second user device **110-2** may take with respect to analytics engine **130**. For example, second user device **110-2** may be allowed to view metrics but not edit them. In some embodiments, the permissions scope may allow second user device **110-2** to edit initiatives and metrics. At **1330**, analytics engine **130** generates an alert comprising a link to access the GUI, wherein the link is unique to the identifier and permission scope associated with the second user device. In some embodiments, the link may be a URL to analytics engine **130**.

[0086] At **1340**, analytics engine **130** sends the alert to the second user device (e.g., user computing device **110-2**). The alert may be sent over network **120**.

[0087] At **1350**, analytics engine **130** receive an interaction, via the GUI, from the second user device. The interaction may be to view the initiative details. In some embodiments, the interaction may be to add additional metrics to the initiative or to edit current initiatives. The interaction may be an action to notify other user computing devices **110** associated with the initiative.

[0088] At **1360**, analytics engine **130** allows the interaction based on a determination that the interaction is allowed by the permission scope associated with the second user device. For example, analytics engine **130** may compare the interaction to the permission scope assigned to the second user device. If the interaction is allowed by the permission scope, analytics engine **130** may allow the interaction to proceed. For example, user computing device **110-1** may add user computing device **110-2** to an initiative and place no restrictions on its permissions. The interaction from user computing device **110-2** may be to add metric associated with the initiative. For example, the initiative may be to reduce surgical waste. The initiative may have a first metric associated with the value of material used during surgery. User computing device **110-2** may add an additional metric corresponding to surgery length. Since user computing device **110-2** has no permission restrictions, analytics engine **130** allows the metric to be added.

[0089] In some embodiments, analytics engine **130** may generate an alert or notification based on the interaction. For example, in the embodiment described above regarding adding a metric, analytics engine **130** may send a notification including the new metric to each user computing device **110**. This may be beneficial so that each user computing device **110** has up to date information regarding the initiative. In some embodiments, if the interaction is outside the permission scope and blocked, analytics engine **130** may generate a notification to notify user computing device **110-1**.

[0090] FIG. **14** is a flowchart illustrating an example process **1400** for using an initiative to identify additional metrics, according to some embodiments. Process **1400** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **14** may be performed by analytics engine **130** and/or user computing device **110**. In some embodiments, one or more of the steps shown in FIG. **14** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **14**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **14** The steps shown in FIG. **14** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **14**.

[0091] At **1410**, analytics engine **130** receives identification of an initiative from user computing device **110**. The initiative may be received via the GUI. The initiative may correspond to a problem or goal of the user or an organization with which the user is associated.

[0092] At **1420**, analytics engine **130** determines, based on an analysis of the initiative, a second metric that is correlated with the initiative. For example, analytics engine **130** may utilize previously received initiatives and metrics associated with those initiatives to identify a second metric. In some embodiments, analytics engine **130** may use artificial intelligence and/or a machine learning model to identify the second metric.

[0093] For example, user computing device **110** may define an initiative to reduce surgical costs and elect to track average surgery length as a metric. Analytics engine **130** may look up an initiative from another user computing device **110** also aimed at reducing surgical costs. In that instance, the other user computing device **110** may have also relied on a metric related to the value of materials used during surgery. Analytics engine **130** may use this correlation to also track the value of materials used during surgery. The correlation may be based on the impact a metric has on an initiative. For example, an initiative aimed at reducing surgical costs may involve thousands of metrics. However, certain metrics may have more direct effects than others. As an example, although the average length of a surgery may increase surgical costs, the value of materials used may have a much greater effect. Therefore, the value of material used may be more important to track than the average length. In some embodiments, analytics engine **130** may send a notification to user computing device **110** that the second metric has been identified. In some embodiments, the user may prevent analytics engine **130** from adding the second metric. For example, the user may remove the second metric from the initiative via the GUI.

[0094] At **1430**, analytics engine **130**, periodically retrieves a second metric value corresponding to the second metric according to an update frequency associated with the second metric, where the update frequency associated with the second metric is greater than an update frequency associated with the first metric. For example, the second metric update frequency may be once per hour and the first metric update frequency may be once per day. The second metric may be updated more frequently than the first based on the determination that the second metric is more closely correlated with the initiative. Using the example above, analytics engine **130** may retrieve a metric for the value of material consumed during surgery more frequently than average surgery length.

[0095] FIG. **15** is a flowchart illustrating an example process **1500** for displaying multiple metrics, according to some embodiments. Process **1500** may be performed after process **1400**, in which a second metric is linked to an initiative. Process **1500** may be performed by processing logic that may comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions executing on a processing device), or a combination thereof. One or more of the steps in FIG. **15** may be performed by analytics engine **130** and/or user computing device **110**. In some embodiments, one or more of the steps shown in FIG. **15** may be omitted, repeated, and/or performed in a different order than the order shown in FIG. **15**. Accordingly, the scope of the invention should not be considered limited to the specific arrangement of steps shown in FIG. **15** The steps shown in FIG. **15** may be implemented as computer-readable instructions stored on computer-readable media, where, when the instructions are executed, cause a processor to perform the process of FIG. **15**.

[0096] At **1510**, analytics engine **130** may customize the GUI by displaying the initiative, the second metric and the second metric value. This customization may be in response to determining that the second metric has a higher correlation or impact on the initiative than the first metric.

[0097] At **1520**, analytics engine **130** may receive an interaction, via the GUI, with the displayed initiative. For example, the user may click the refresh button (e.g., refresh **365** of FIG. **3**) or view the details button (e.g., view details **350** of FIG. **3**).

[0098] At **1530**, in response to the interaction, analytics engine **130** may retrieve updated first and second metric values. The updated first and second metric values may be retrieved from data provider systems **140**.

[0099] At **1540**, analytics engine **130** stores the updated first and second metric values at the first local data store. The updated first and second metric values may be stored at local store **134**. This is beneficial to ensure that user computing device **110** has rapid access to the updated metric values.

[0100] At **1540**, analytics engine **130** may update the GUI to display the initiative, the first metric, the updated first metric value, the second metric and the updated second metric value. This operation is beneficial to allow user computing device **110** to display both metrics tied to the initiative.

[0101] Various embodiments may be implemented, for example, using one or more well-known computer systems, such as computer system **1600** shown in FIG. **16**. One or more computer systems **1600** may be used, for example, to implement any of the embodiments discussed herein, as well as combinations and sub-combinations thereof.

[0102] Computer system **1600** may include one or more processors (also called central processing units, or CPUs), such as a processor **1604**. Processor **1604** may be connected to a communication infrastructure or bus **1606**.

[0103] Computer system **1600** may also include user input/output device(s) **1603**, such as monitors, keyboards, pointing devices, etc., which may communicate with communication infrastructure **1606** through user input/output interface(s) **1602**.

[0104] One or more of processors **1604** may be a graphics processing unit (GPU). In an embodiment, a GPU may be a processor that is a specialized electronic circuit designed to process mathematically intensive applications. The GPU may have a parallel structure that is efficient for parallel processing of large blocks of data, such as mathematically intensive data common to computer graphics applications, images, videos, etc.

[0105] Computer system **1600** may also include a main or primary memory **1608**, such as random access memory (RAM). Main memory **1608** may include one or more levels of cache. Main memory **1608** may have stored therein control logic (i.e., computer software) and/or data.

[0106] Computer system **1600** may also include one or more secondary storage devices or memory **1610**. Secondary memory **1610** may include, for example, a hard disk drive **1612** and/or a removable storage device or drive **1614**. Removable storage drive **1614** may be a floppy disk drive, a magnetic tape drive, a compact disk drive, an optical storage device, tape backup device, and/or any other storage device/drive.

[0107] Removable storage drive **1614** may interact with a removable storage unit **1618**. Removable storage unit **1618** may include a computer usable or readable storage device having stored thereon computer software (control logic) and/or data. Removable storage unit **1618** may be a floppy disk, magnetic tape, compact disk, DVD, optical storage disk, and/any other computer data storage device. Removable storage drive **1614** may read from and/or write to removable storage unit **1618**.

[0108] Secondary memory **1610** may include other means, devices, components, instrumentalities or other approaches for allowing computer programs and/or other instructions and/or data to be accessed by computer system **1600**. Such means, devices, components, instrumentalities or other approaches may include, for example, a removable storage unit **1622** and an interface **1620**. Examples of the removable storage unit **1622** and the interface **1620** may include a program cartridge and cartridge interface (such as that found in video game devices), a removable memory chip (such as an EPROM or PROM) and associated socket, a memory stick and USB port, a memory card and associated memory card slot, and/or any other removable storage unit and associated interface.

[0109] Computer system **1600** may further include a communication or network interface **1624**. Communication interface **1624** may enable computer system **1600** to communicate and interact with any combination of external devices, external networks, external entities, etc. (individually and collectively referenced by reference number **1628**). For example, communication interface **1624** may allow computer system **1600** to communicate with external or remote devices **1628** over communications path **1626**, which may be wired and/or wireless (or a combination thereof), and which may include any combination of LANs, WANs, the Internet, etc. Control logic and/or data may be transmitted to and from computer system **1600** via communication path **1626**.

[0110] Computer system **1600** may also be any of a personal digital assistant (PDA), desktop workstation, laptop or notebook computer, netbook, tablet, smart phone, smart watch or other wearable, appliance, part of the Internet-of-Things, and/or embedded system, to name a few non-limiting examples, or any combination thereof.

[0111] Computer system **1600** may be a client or server, accessing or hosting any applications and/or data through any delivery paradigm, including but not limited to remote or distributed cloud computing solutions; local or on-premises software ("on-premise" cloud-based solutions); "as a service" models (e.g., content as a service (CaaS), digital content as a service (DCaaS), software as a service (SaaS), managed software as a service (MSaaS), platform as a service (PaaS), desktop as a service (DaaS), framework as a service (FaaS), backend as a service (BaaS), mobile backend as a service (MBaaS), infrastructure as a service (IaaS), etc.); and/or a hybrid model including any combination of the foregoing examples or other services or delivery paradigms.

[0112] Any applicable data structures, file formats, and schemas in computer system **1600** may be derived from standards including but not limited to JavaScript Object Notation (JSON), Extensible Markup Language (XML), Yet Another Markup Language (YAML), Extensible Hypertext Markup Language (XHTML), Wireless Markup Language (WML), MessagePack, XML User Interface

Language (XUL), or any other functionally similar representations alone or in combination. Alternatively, proprietary data structures, formats or schemas may be used, either exclusively or in combination with known or open standards.

[0113] In some embodiments, a tangible, non-transitory apparatus or article of manufacture comprising a tangible, non-transitory computer useable or readable medium having control logic (software) stored thereon may also be referred to herein as a computer program product or program storage device. This includes, but is not limited to, computer system **1600**, main memory **1608**, secondary memory **1610**, and removable storage units **1618** and **1622**, as well as tangible articles of manufacture embodying any combination of the foregoing. Such control logic, when executed by one or more data processing devices (such as computer system **1600**), may cause such data processing devices to operate as described herein.

[0114] Based on the teachings contained in this disclosure, it will be apparent to persons skilled in the relevant art(s) how to make and use embodiments of this disclosure using data processing devices, computer systems and/or computer architectures other than that shown in FIG. **16**. In particular, embodiments can operate with software, hardware, and/or operating system implementations other than those described herein.

[0115] It is to be appreciated that the Detailed Description section, and not any other section, is intended to be used to interpret the claims. Other sections can set forth one or more but not all exemplary embodiments as contemplated by the inventor(s), and thus, are not intended to limit this disclosure or the appended claims in any way.

[0116] While this disclosure describes exemplary embodiments for exemplary fields and applications, it should be understood that the disclosure is not limited thereto. Other embodiments and modifications thereto are possible, and are within the scope and spirit of this disclosure. For example, and without limiting the generality of this paragraph, embodiments are not limited to the software, hardware, firmware, and/or entities illustrated in the figures and/or described herein. Further, embodiments (whether or not explicitly described herein) have significant utility to fields and applications beyond the examples described herein.

[0117] Embodiments have been described herein with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined as long as the specified functions and relationships (or equivalents thereof) are appropriately performed. Also, alternative embodiments can perform functional blocks, steps, operations, methods, etc. using orderings different than those described herein.

[0118] References herein to “one embodiment,” “an embodiment,” “an example embodiment,” or similar phrases, indicate that the embodiment described can include a particular feature, structure, or characteristic, but every embodiment can not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it would be within the knowledge of persons skilled in the relevant art(s) to incorporate such feature, structure, or characteristic into other embodiments whether or not explicitly mentioned or described herein. Additionally, some embodiments can be described using the expression “coupled” and “connected” along with their derivatives. These terms are not necessarily intended as synonyms for each other. For example, some embodiments can be described using the terms “connected” and/or “coupled” to indicate that two or more elements are in direct physical or electrical contact with each other. The term “coupled,” however, can also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other.

[0119] The breadth and scope of this disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A computer-implemented method, comprising: generating a graphical user interface (GUI) in response to a request from a user device, wherein the request includes a location of the user device; receiving, via the GUI, an interaction comprising an identification of an initiative, wherein the identification of the initiative is a text-based description; identifying a first metric having a first metric value and correlated with the initiative by: determining the initiative is similar to a previously-received initiative based on a comparison of the text-based description of the initiative to a text-based description of the previously-received initiative; and analyzing the previously-received initiative to determine that a previous change in the first metric value was correlated with a change in status of the previously-received initiative; identifying a data source comprising the first metric, wherein the data source stores a plurality of metrics; retrieving the first metric value of the first metric from the data source; determining a first local data store that is located closer in proximity to the location of the user device than the data source; storing the first metric and the first metric value at the first local data store such that the first metric and the first metric value are stored closer in proximity to the user device than the plurality of metrics stored at the data source; and generating an updated GUI displaying the initiative, the first metric, and the first metric value, wherein the GUI is updated based on retrieving the first metric and the first metric value from the first local data store.
2. (canceled)
3. The computer-implemented method of claim 1, further comprising: detecting that the user device moved to a new location; identifying a second local data store that is closer in proximity to the new location than the first local data store; and storing the first metric value at the second local data store.
4. The computer-implemented method of claim 1, further comprising: receiving a target metric value corresponding to the first metric; calculating a difference between the target metric value and the first metric value; generating a report including the first metric, the first metric value, the target metric value, and the difference between the target metric value and the first metric value; sending the report to the user device; and displaying the report within the GUI.
5. The computer-implemented method of claim 1, further comprising: receiving, via the GUI, an identifier associated with a second user device; assigning a permission scope to the second user device, wherein the permission scope is configured to block an interaction from the second user device with the GUI; generating an alert comprising a link to access the second GUI, wherein the link is unique to the identifier and the permission scope associated with the second user device, sending the alert to the second user device; receiving, via the GUI, an interaction from the second user device; and allowing the interaction based on a determination that the interaction is allowed by the permission scope associated with the second user device.
6. The computer-implemented method of claim 1, further comprising: determining, based on an analysis of the initiative, a second metric that is correlated with the initiative; and periodically retrieving a second metric value corresponding to the second metric according to an update frequency associated with the second metric, wherein the update frequency associated with the second metric is greater than an update frequency associated with the first metric.
7. The computer-implemented method of claim 6, further comprising: customizing the GUI by displaying the initiative, the second metric and the second metric value; receiving an interaction, via the GUI, with the displayed initiative; in response to receiving the interaction, retrieving updated first and second metric values; storing the updated first and second metric values at the first local data store; and updating the GUI to display the initiative, the first metric, the updated

first metric value, the second metric and the updated second metric value.

8. A system, comprising: a memory; and at least one processor coupled to the memory and configured to: generate a graphical user interface (GUI) in response to a request from a user device, wherein the request includes a location of the user device; receive, via the GUI, an interaction comprising an identification of an initiative, wherein the identification of the initiative is a text-based description; identify a first metric having a first metric value and correlated with the initiative by: determining the initiative is similar to a previously-received initiative based on a comparison of the text-based description of the initiative to a text-based description of the previously-received initiative; and analyzing the previously-received initiative to determine that a previous change in the first metric value was correlated with a change in status of the previously-received initiative; identify a data source comprising the first metric, wherein the data source stores a plurality of metrics;; retrieve the first metric value of the first metric from the data source; determine a first local data store that is located closer in proximity to the location of the user device than the data source; store the first metric and the first metric value at the first local data store such that the first metric and the first metric value are stored closer in proximity to the user device than the plurality of metrics stored at the data source; and generate an updated GUI displaying the initiative, the first metric, and the first metric value, wherein the GUI is updated based on retrieving the first metric and the first metric value from the first local data store.

9. The system of claim 8, wherein the at least one processor is further configured to: configure a batch process to retrieve the first metric value, the batch process comprising: an update frequency; and the first metric; execute the batch process according to the update frequency; and update the first metric value at the first local data store.

10. The system of claim 8, wherein the at least one processor is further configured to: detect that the user device moved to a new location; identify a second local data store that is located closer in proximity to the new location than the first local data store; and store the first metric value at the second local data store.

11. The system of claim 8, wherein the at least one processor is further configured to: receive a target metric value corresponding to the first metric; calculate a difference between the target metric value and the first metric value; generate a report including the first metric, the first metric value, the target metric value, and the difference between the target metric value and the first metric value; send the report to the user device; and display the report within the GUI.

12. The system of claim 8, wherein the at least one processor is further configured to: receive, via the GUI, an identifier associated with a second user device; assign a permission scope to second user device, wherein the permission scope is configured to block an interaction from the second user device with the GUI; generate an alert comprising a link to access the second GUI, wherein the link is unique to the identifier and the permission scope associated with the second user device, send the alert to the second user device; receive, via the GUI, an interaction from the second user device; and allow the interaction based on a determination that the interaction is allowed by the permission scope associated with the second user device.

13. The system of claim 8, wherein the at least one processor is further configured to: determine, based on an analysis of the initiative, a second metric that is correlated with the initiative; and periodically retrieve a second metric value corresponding to the second metric according to an update frequency associated with the second metric, wherein the update frequency associated with the second metric is greater than an update frequency associated with the first metric.

14. The system of claim 13, wherein the at least one processor is further configured to: customize the GUI by displaying the initiative, the second metric and the second metric value; receive an interaction, via the GUI, with the displayed initiative; in response to receiving the interaction, retrieve updated first and second metric values; store the updated first and second metric values at the first local data store; and update the GUI to display the initiative, the first metric, the updated first metric value, the second metric and the updated second metric value.

15. A non-transitory computer-readable device having instructions stored thereon that, when executed by at least one computing device, cause the at least one computing device to perform operations comprising: generating a graphical user interface (GUI) in response to a request from a user device, wherein the request includes a location of the user device; receiving, via the GUI, an interaction comprising an identification of an initiative, wherein the identification of the initiative is a text-based description; identifying a first metric having a first metric value and correlated with the initiative by: determining the initiative is similar to a previously-received initiative based on a comparison of the text-based description of the initiative to a text-based description of the previously-received initiative; and analyzing the previously-received initiative to determine that a previous change in the first metric value was correlated with a change in status of the previously-received initiative; identifying a data source comprising the first metric, wherein the data source stores a plurality of metrics; retrieving the first metric value of the first metric from the data source; determining a first local data store that is located closer in proximity to the location of the user device than the data source; storing the first metric and the first metric value at the first local data store such that the first metric and the first metric value are stored closer in proximity to the user device than the plurality of metrics stored at the data source; and generating an updated GUI displaying the initiative, the first metric, and the first metric value, wherein the GUI is updated based on retrieving the first metric and the first metric value from the first local data store.
16. The non-transitory computer-readable device of claim 15, the operations further comprising: configuring a batch process to retrieve the first metric value, the batch process comprising; an update frequency; and the first metric; executing the batch process according to the update frequency; and updating the first metric value at the first local data store.
17. The non-transitory computer-readable device of claim 15, the operations further comprising: detecting that the user device moved to a new location; identifying a second local data store that is located closer in proximity to the new location than the first local data store; and storing the first metric value at the second local data store.
18. The non-transitory computer-readable device of claim 15, the operations further comprising: receiving a target metric value corresponding to the first metric; calculating a difference between the target metric value and the first metric value; generating a report including the first metric, the first metric value, the target metric value, and the difference between the target metric value and the first metric value; sending the report to the user device; and displaying the report within the GUI.
19. The non-transitory computer-readable device of claim 15, the operations further comprising: receiving, via the GUI, an identifier associated with a second user device; assigning a permission scope to the second user device, wherein the permission scope is configured to block an interaction from the second user device with the GUI; generating an alert comprising a link to access the second GUI, wherein the link is unique to the identifier and the permission scope associated with the second user device, sending the alert to the second user device; receiving, via the GUI, an interaction from the second user device; and allowing the interaction based on a determination that the interaction is allowed by the permission scope associated with the second user device.
20. The non-transitory computer-readable device of claim 15, the operations further comprising: determining, based on an analysis of the initiative, a second metric that is correlated with the initiative; and periodically retrieving a second metric value corresponding to the second metric according to an update frequency associated with the second metric, wherein the update frequency associated with the second metric is greater than an update frequency associated with the first metric.
21. The computer-implemented method of claim 1, wherein the first metric is identified using a machine learning model, and wherein training the machine learning model comprises: generating a list of previously received initiatives, each previously received initiative including a description, a metric, a target metric value, and a time taken to reach the target metric value, wherein the list of previously received initiatives includes a first initiative and a second initiative that share a

description, and wherein the time taken to reach the target metric value for the first initiative is less than the time taken to reach the target metric value for the second initiative; and training the machine learning model using the list of previously received initiatives, such that the machine learning model is configured to predict the metric of the first initiative when a new initiative with a description matching the description of the first initiative is received.